

Vo Hong Quan

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Education

Ho Chi Minh City University of Technology and Engineering, Ho Chi Minh City, Vietnam Expected Dec 2026
Bachelor of Robotics and Artificial Intelligence GPA: 3.22/4.0
Relevant coursework: Machine Vision, Artificial Intelligence, Artificial Neural Networks, Practice of Artificial Intelligence

Experience

Robotics Lab, Ho Chi Minh City University of Technology and Engineering Feb 2025 – Present
Research Assistant Ho Chi Minh City, Vietnam

- Support reinforcement learning training and real-time testing of robot control systems.
- Assist in designing reward functions and tuning hyperparameters for reinforcement learning agents.

Projects

Crazyflie-IsaacLab: RL-Tuned PID Gains for a Crazyflie Quadrotor Jan 2026 – Sep 2026
Python, PyTorch, PPO (rsl_rl), NVIDIA Isaac Lab github.com/vohongquann/Crazyflie-IsaacLab

- Built a four-layer cascaded flight controller (position 50 Hz, velocity 100 Hz, attitude 250 Hz, rate 500 Hz) for a Crazyflie 2.1 Brushless in Isaac Lab 3.0, with physical constants traced to the datasheet, system-identification papers and Bitcraze firmware.
- Trained one small MLP per layer with PPO, bottom-up and then frozen, to output the 9 PID gains of its layer at every step; cut tracking error by 23–62% against the hand-tuned PID on all four layers (attitude 15.5° to 5.9°).
- Designed the gain network so that zero output equals the tuned PID, so training starts from a stable controller and converges in a few hundred PPO iterations; kept the networks small enough for the drone's STM32 microcontroller.
- Measured the cost of the gains: motor command chatter is $2.6\text{--}6.3\times$ the PID's on the outer layers, documented as the next improvement.
- Wrote a simulator-free pytest suite and scripts that regenerate every step-response figure and metric in the report.

BalancingRobot-IsaacLab: Self-Balancing Robot with an RL Cascade Jun 2025 – Dec 2025
Python, PPO (rsl_rl), ROS 2, ONNX, LQR, PID github.com/vohongquann/BalancingRobot-IsaacLab

- Trained a velocity policy with PPO that turns a forward-speed and turn-rate command into wheel torques for a Pololu Balboa 32U4, working only from a raw IMU (complementary filter) and the wheel encoders, as the real robot would.
- Froze it and trained a position policy on top that drives to a goal pose; on 768 randomized episodes (pushes, IMU noise, randomized mass, friction and centre of mass) neither fell, with median errors of 8 mm (position) and 0.021 m/s (velocity).
- Benchmarked against LQR and PID on the same tests: speed-step error 0.026 m/s settled in 0.36 s, versus 0.067 m/s and 1.1 s for LQR; goals reached within 2–6 mm, versus 13–50 mm for LQR.
- Trained policies that tune PID gains instead of writing torques; the gain-tuned go-to-goal PID followed a figure 8 with a 0.043 m median distance, the best of the four controllers.
- Deployed the same networks (ONNX) in a ROS 2 node that reads the raw IMU and joint states and publishes wheel torques.

RotaryPendulum-IsaacLab: Learned Cascade PID for a Rotary Inverted Pendulum Feb 2025 – May 2025
Python, PPO (rsl_rl), NVIDIA Isaac Lab, PhysX, URDF github.com/vohongquann/RotaryPendulum-IsaacLab

- Modeled the QUBE-Servo 2 in Isaac Lab: URDF generated from CAD, a DC-motor model (PWM, current, back-EMF, torque) and quantized encoders with filtered velocity, simulated at 500 Hz.
- Trained a PPO policy (4,096 parallel robots, 1,500 iterations, 64 minutes on one RTX 3060) that outputs the six gains of a two-loop cascade PID at 100 Hz from 12 observations.
- Validated on 256 simulated robots, each with its own motor constants: swing-up in 0.40 s (median), arm steps settled in 0.23–0.50 s with at most 1.4° median overshoot (3 of 256 episodes ended early).
- Published the trained weights, a 1080p demo video and the evaluation scripts so a fresh clone plays and re-evaluates in one command.

Skills

Programming: Python, C/C++, MATLAB (Simulink, Simscape, RL Toolbox)

Reinforcement Learning & Control: PPO, SAC, reward design, hyperparameter tuning, cascaded PID, LQR

Robotics & Simulation: NVIDIA Isaac Sim, Isaac Lab, PhysX, ROS 2, URDF, ESP32

Machine Learning & Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLO

Tools: Git, GitHub, Linux, Conda, Jupyter/Colab, pytest

Languages: Vietnamese (native), English (TOEIC 715/990)

Certifications

TOEIC Listening & Reading: 715/990, IIG Vietnam

2025